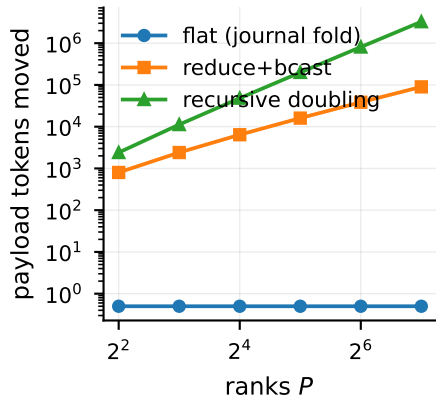
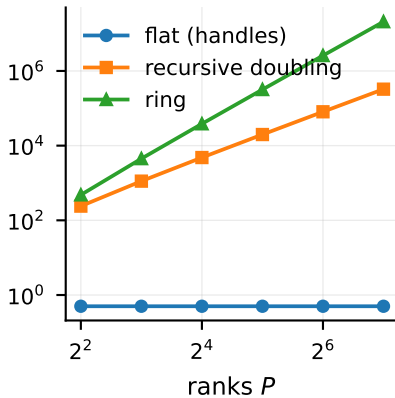


(a) allreduce



(b) allgather



(c) barrier

